CSE-ARTIFICIAL INTELLIGENCE &MACHINE LEARNING

PROJECT ON

DISPLAYING STUDENT DETAILS USING PATTERNS

STUDENT BY:

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CLASS: **III – AIML – A**

SUBMITTED TO:

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AIM:
To **print custom text banners** using characters made from asterisks (*) in **5 x 5 matrix patterns**. The banner represents:

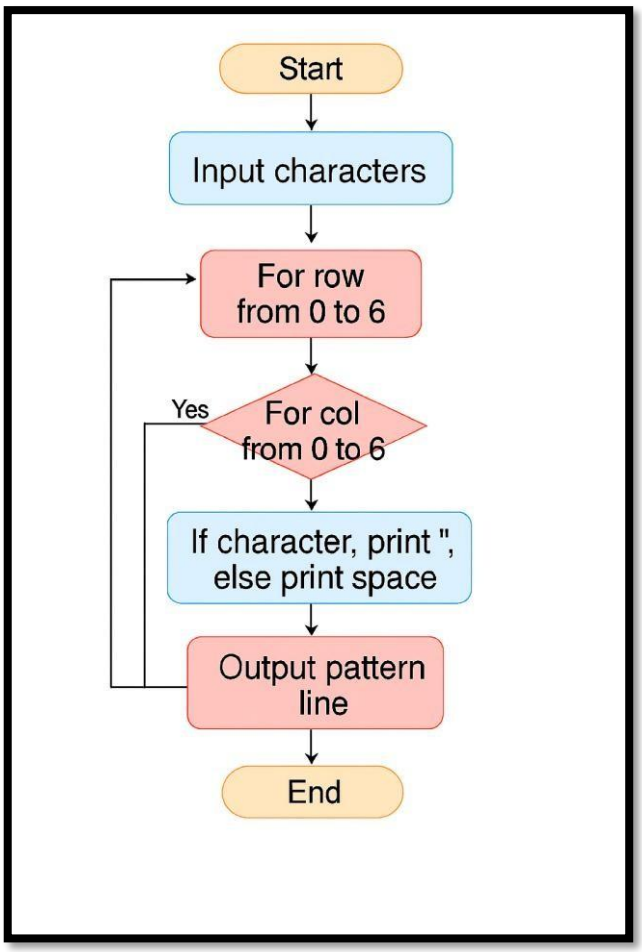
- A **name** (e.g., NKUSARAJA)
- A **roll number** (e.g., 23P31A4252)
- A **branch** (e.g., AIML)
- A **college** (e.g., ACET)

Each character (A-Z, 0-9) is drawn manually using logic that determines which cells in a 5x5 matrix should be filled with *.

SOFTWARE REQUIREMENTS:

Component	**Description**
Language	Python
Python- Version	Python – 3.13.1
Code Editor	VS Code
Execution	Terminal

FLOW CHART: -



ALGORITHM:

1. **Start Program**
2. **Initialize Variables:**
    - Define constant n = 5 (for the 5x5 pattern size).
    - Define required variables (i, j) for looping.
    - Define character arrays for:
    - Name
    - Roll Number
    - Branch
    - College
3. **For Each Input String:**

   Handle each input line separately (Name, Roll Number, Branch, College).
4. **For Each Row of the Pattern (1 to n):**
    - Loop over all 5 rows for the current string.

5. **For Each Character in the String:**

- For each character in the string, print its pattern for the current row:
- If Character is a Letter or Digit:
- Use if-else or switch-case to determine the pattern for each character.
- Print * or spaces based on the pattern logic for that character and current row.
- If Character is a Space:
- Print extra spaces to separate words.

6. **End Current Row**

- Print a newline after finishing all characters in the current row.

7. **Add Vertical Spacing Between Lines**

- After printing one full input string (all rows), print one or two blank lines before starting the next set (for Roll, Branch, or College).

8. **Repeat Steps 4–7 for All Strings**

- Do the row-by-row pattern printing for Name, Roll Number, Branch, and College separately.

9. **End Program**

PROGRAM CODE:

```
# Python implementation to print the
# pattern of alphabets A to Z using *, row by row for a string

height = 5
width = (2 * height) - 1

def abs_val(d):
    return -d if d < 0 else d

def printA(ind):
    n = width // 2
    for i in range(height):
        if i == ind:
            for j in range(width + 1):
                if j == n or j == (width - n) or (i == height // 2 and j > n and j < (width
- n)):
                    print('*', end='')
```

```
            else:
                print(' ', end='')
        n -= 1

def printB(ind):
    half = height // 2
    for i in range(height):
        if i == ind:
            print('*', end='')
            for j in range(width):
                if (i == 0 or i == height - 1 or i == half) and j < (width - 2):
                    print('*', end='')
                elif j == (width - 2) and not (i == 0 or i == height - 1 or i == half):
                    print('*', end='')
                else:
                    print(' ', end='')

def printC(ind):
    for i in range(height):
        if i == ind:
            print('*', end='')
            for j in range(height - 1):
                if i == 0 or i == height - 1:
                    print('*', end='')
                else:
                    print(' ', end='')

def printD(ind):
    for i in range(height):
        if i == ind:
            print('*', end='')
            for j in range(height):
                if (i == 0 or i == height - 1) and j < height - 1:
                    print('*', end='')
                elif j == height - 1 and i != 0 and i != height - 1:
                    print('*', end='')
                else:
                    print(' ', end='')

def printE(ind):
```

```
    for i in range(height):
        if i == ind:
            print('*', end='')
            for j in range(height):
                if (i == 0 or i == height - 1) or (i == height // 2 and j <= height // 2):
                    print('*', end='')
                else:
                    print(' ', end='')

def printF(ind):
    for i in range(height):
        if i == ind:
            print('*', end='')
            for j in range(height):
                if (i == 0) or (i == height // 2 and j <= height // 2):
                    print('*', end='')
                else:
                    print(' ', end='')

def printG(ind):
    local_width = width - 1
    for i in range(height):
        if i == ind:
            for j in range(local_width):
                if (i == 0 or i == height - 1) and (j == 0 or j == local_width - 1):
                    print(' ', end='')
                elif j == 0:
                    print('*', end='')
                elif i == 0 and j <= height:
                    print('*', end='')
                elif i == height // 2 and j > height // 2:
                    print('*', end='')
                elif i > height // 2 and j == local_width - 1:
                    print('*', end='')
                elif i == height - 1 and j < local_width:
                    print('*', end='')
                else:
                    print(' ', end='')

def printH(ind):
```

```
    for i in range(height):
        if i == ind:
            print('*', end='')
            for j in range(height):
                if j == height - 1 or i == height // 2:
                    print('*', end='')
                else:
                    print(' ', end='')

def printI(ind):
    for i in range(height):
        if i == ind:
            for j in range(height):
                if i == 0 or i == height - 1:
                    print('*', end='')
                elif j == height // 2:
                    print('*', end='')
                else:
                    print(' ', end='')

def printJ(ind):
    for i in range(height):
        if i == ind:
            for j in range(height):
                if i == height - 1 and (j > 0 and j < height - 1):
                    print('*', end='')
                elif (j == height - 1 and i != height - 1) or (i > (height // 2) - 1 and j ==
0 and i != height - 1):
                    print('*', end='')
                else:
                    print(' ', end='')

def printK(ind):
    half = height // 2
    dummy = half
    for i in range(height):
        if i == ind:
            print('*', end='')
            for j in range(half + 1):
                if j == abs_val(dummy):
```

```
                print('*', end='')
            else:
                print(' ', end='')
        dummy -= 1

def printL(ind):
    for i in range(height):
        if i == ind:
            print('*', end='')
            for j in range(height + 1):
                if i == height - 1:
                    print('*', end='')
                else:
                    print(' ', end='')

def printM(ind):
    counter = 0
    for i in range(height):
        if i == ind:
            print('*', end='')
            for j in range(height + 1):
                if j == height:
                    print('*', end='')
                elif j == counter or j == height - counter - 1:
                    print('*', end='')
                else:
                    print(' ', end='')
        if counter == height // 2:
            counter = -99999
        else:
            counter += 1

def printN(ind):
    counter = 0
    for i in range(height):
        if i == ind:
            print('*', end='')
            for j in range(height + 1):
                if j == height:
                    print('*', end='')
```

```
            elif j == counter:
                print('*', end='')
            else:
                print(' ', end='')
        counter += 1

def printO(ind):
    space = (height // 3)
    local_width = height // 2 + height // 5 + space + space
    sp = space
    for i in range(height):
        if i == ind:
            for j in range(local_width + 1):
                if j == local_width - abs_val(sp) or j == abs_val(sp):
                    print('*', end='')
                elif (i == 0 or i == height - 1) and j > abs_val(sp) and j < local_width -
abs_val(sp):
                    print('*', end='')
                else:
                    print(' ', end='')
        if sp != 0 and i < height // 2:
            sp -= 1
        elif i >= (height // 2 + height // 5):
            sp -= 1

def printP(ind):
    for i in range(height):
        if i == ind:
            print('*', end='')
            for j in range(height):
                if (i == 0 or i == height // 2) and j < height - 1:
                    print('*', end='')
                elif i < height // 2 and j == height - 1 and i != 0:
                    print('*', end='')
                else:
                    print(' ', end='')

def printQ(ind):
    space = (height // 3)
    local_width = height // 2 + height // 5 + space + space
```

```
    sp = space
    for i in range(height):
        if i == ind:
            for j in range(local_width + 1):
                if j == local_width - abs_val(sp) or j == abs_val(sp):
                    print('*', end='')
                elif (i == 0 or i == height - 1) and j > abs_val(sp) and j < local_width -
abs_val(sp):
                    print('*', end='')
                else:
                    print(' ', end='')
        if sp != 0 and i < height // 2:
            sp -= 1
        elif i >= (height // 2 + height // 5):
            sp -= 1
    d = height
    for i in range(height // 2):
        if height + i == ind:
            for j in range(d + 1):
                if j == d:
                    print('*', end='')
                else:
                    print(' ', end='')
            d += 1

def printR(ind):
    half = height // 2
    for i in range(height):
        if i == ind:
            print('*', end='')
            for j in range(width):
                if (i == 0 or i == half) and j < (width - 2):
                    print('*', end='')
                elif j == (width - 2) and not (i == 0 or i == half):
                    print('*', end='')
                else:
                    print(' ', end='')

def printS(ind):
    for i in range(height):
```

```
        if i == ind:
            for j in range(height):
                if i == 0 or i == height // 2 or i == height - 1:
                    print('*', end='')
                elif i < height // 2 and j == 0:
                    print('*', end='')
                elif i > height // 2 and j == height - 1:
                    print('*', end='')
                else:
                    print(' ', end='')

def printT(ind):
    for i in range(height):
        if i == ind:
            for j in range(height):
                if i == 0:
                    print('*', end='')
                elif j == height // 2:
                    print('*', end='')
                else:
                    print(' ', end='')

def printU(ind):
    for i in range(height):
        if i == ind:
            if i != 0 and i != height - 1:
                print('*', end='')
            else:
                print(' ', end='')
            for j in range(height):
                if i == height - 1 and 0 <= j < height - 1:
                    print('*', end='')
                elif j == height - 1 and i != 0 and i != height - 1:
                    print('*', end='')
                else:
                    print(' ', end='')

def printV(ind):
    counter = 0
    for i in range(height):
```

```
        if i == ind:
            for j in range(width + 1):
                if j == counter or j == width - counter:
                    print('*', end='')
                else:
                    print(' ', end='')
        counter += 1

def printW(ind):
    counter = height // 2
    for i in range(height):
        if i == ind:
            print('*', end='')
            for j in range(height + 1):
                if j == height:
                    print('*', end='')
                elif i >= height // 2 and (j == counter or j == height - counter - 1):
                    print('*', end='')
                else:
                    print(' ', end='')
        if i >= height // 2:
            counter += 1

def printX(ind):
    counter = 0
    for i in range(height + 1):
        if i == ind:
            for j in range(height + 1):
                if j == counter or j == height - counter:
                    print('*', end='')
                else:
                    print(' ', end='')
        counter += 1

def printY(ind):
    counter = 0
    for i in range(height):
        if i == ind:
            for j in range(height + 1):
                if j == counter or (j == height - counter and i <= height // 2):
```

```
                print('*', end='')
            else:
                print(' ', end='')
        if i < height // 2:
            counter += 1

def printZ(ind):
    counter = height - 1
    for i in range(height):
        counter -= 1
        if i == ind:
            for j in range(height):
                if i == 0 or i == height - 1 or j == counter:
                    print('*', end='')
                else:
                    print(' ', end='')
def print0(ind):
    for i in range(height):
        if i == ind:
            for j in range(height):
                if ((j == 0 or j == height - 1) and i != 0 and i != height - 1) or ((i == 0
or i == height - 1) and j > 0 and j < height - 1):
                    print('*', end='')
                else:
                    print(' ', end='')

def print1(ind):
    for i in range(height):
        if i == ind:
            for j in range(height):
                if j == height // 2 or (i == height - 1 and j > 0 and j < height - 1):
                    print('*', end='')
                else:
                    print(' ', end='')

def print2(ind):
    for i in range(height):
        if i == ind:
            for j in range(height):
                if (i == 0 or i == height - 1 or i == height // 2):
```

```
                print('*', end='')
            elif i < height // 2 and j == height - 1:
                print('*', end='')
            elif i > height // 2 and j == 0:
                print('*', end='')
            else:
                print(' ', end='')

def print3(ind):
    for i in range(height):
        if i == ind:
            for j in range(height):
                if (i == 0 or i == height - 1 or i == height // 2):
                    print('*', end='')
                elif j == height - 1:
                    print('*', end='')
                else:
                    print(' ', end='')

def print4(ind):
    for i in range(height):
        if i == ind:
            for j in range(height):
                if (j == height - 1 or i == height // 2):
                    print('*', end='')
                elif i < height // 2 and j == 0:
                    print('*', end='')
                else:
                    print(' ', end='')

def print5(ind):
    for i in range(height):
        if i == ind:
            for j in range(height):
                if (i == 0 or i == height - 1 or i == height // 2):
                    print('*', end='')
                elif i < height // 2 and j == 0:
                    print('*', end='')
                elif i > height // 2 and j == height - 1:
                    print('*', end='')
```

```
            else:
                print(' ', end='')

def print6(ind):
    for i in range(height):
        if i == ind:
            for j in range(height):
                if (i == 0 or i == height - 1 or i == height // 2):
                    print('*', end='')
                elif (i < height // 2 and j == 0) or (i > height // 2 and (j == 0 or j ==
height - 1)):
                    print('*', end='')
                elif i == height // 2 and j == height - 1:
                    print('*', end='')
                else:
                    print(' ', end='')

def print7(ind):
    for i in range(height):
        if i == ind:
            for j in range(height):
                if i == 0:
                    print('*', end='')
                elif j == height - 1 - i:
                    print('*', end='')
                else:
                    print(' ', end='')

def print8(ind):
    for i in range(height):
        if i == ind:
            for j in range(height):
                if (i == 0 or i == height - 1 or i == height // 2):
                    print('*', end='')
                elif j == 0 or j == height - 1:
                    print('*', end='')
                else:
                    print(' ', end='')

def print9(ind):
```

```
    for i in range(height):
        if i == ind:
            for j in range(height):
                if (i == 0 or i == height // 2 or i == height - 1):
                    print('*', end='')
                elif i < height // 2 and (j == 0 or j == height - 1):
                    print('*', end='')
                elif i > height // 2 and j == height - 1:
                    print('*', end='')
                else:
                    print(' ', end='')
def printPattern(str):
    pattern_functions = {
        'A': printA, 'B': printB, 'C': printC, 'D': printD, 'E': printE,
        'F': printF, 'G': printG, 'H': printH, 'I': printI, 'J': printJ,
        'K': printK, 'L': printL, 'M': printM, 'N': printN, 'O': printO,
        'P': printP, 'Q': printQ, 'R': printR, 'S': printS, 'T': printT,
        'U': printU, 'V': printV, 'W': printW, 'X': printX, 'Y': printY, 'Z': printZ,
        '0': print0, '1': print1, '2': print2, '3': print3, '4': print4,
        '5': print5, '6': print6, '7': print7, '8': print8, '9': print9
    }
    for i in range(5):
        for j in str:
            pattern_functions[j](i)
            print('  ',end = '')
        print()

if __name__ == '__main__':
    name = “NKUSARAJA”
    rollno = “23P31A4252”
    branch = “AIML”
    clg = “ACET”
    printPattern(name)
    printPattern(rollno)
    printPattern(branch)
    printPattern(clg)
```

OUTPUT:

```
**     *  *  *          *****       **       ********         **           *       **
* *    *  * *   *     *  *          *  *      *        *       *  *          *      *  *
*  *   *  **    *     *  *****    ******    ********      ******    *    *    ******
*   * *  * *   *     *      *    *       *   *        *    *       *   *    *    *       *
*    **  *  *   ****   *****  *         *  *        *   *         *   ***   *         *
*****  *****  *****   *****    *         **      *   *  *****  *****  *****
     *      *  *    *      *    *        *  *     *   *      *  *          *
*****  *****  *****   *****    *      ******    *****  *****  *****  *****
*          *  *           *    *     *      *       *  *          *  *
*****  *****  *       *****   ***   *        *      *  *****  *****  *****
     **      *****  **   **  *
    *  *       *    * * * *  *
   ******      *    *  *  *  *
  *      *     *    *     *  *
*        *  *****  *     *  *******
     **      *****  ******  *****
    *  *     *      *         *
   ******    *      ****      *
  *      *   *      *         *
*        *  *****  ******    *
```